



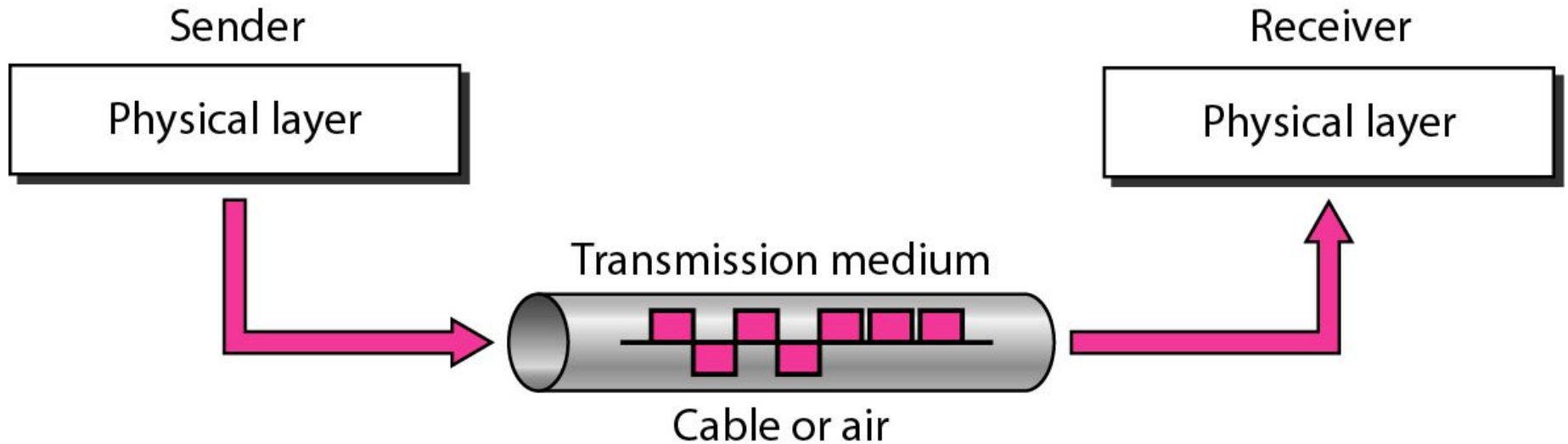
Module - II

PHYSICAL LAYER

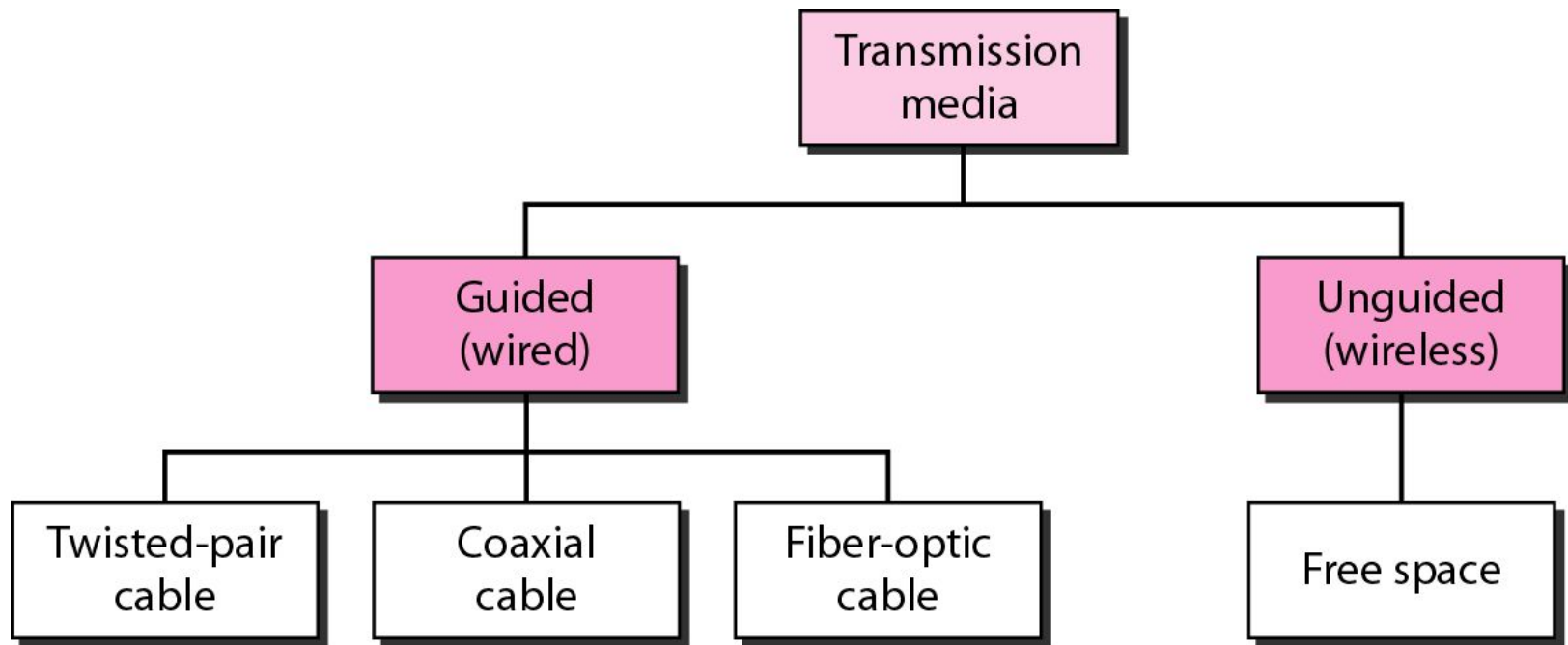
CONTENTS:

- Physical layer:
 - Guided and unguided transmission media, Data transmission
- Data link layer:
 - Introduction to data link layer, framing, Error detection and correction techniques, data link layer protocols,
 - multiple access:
 - random access: ALOHA, Carrier Sense Multiple Access (CSMA), Carrier Sense Multiple Access with Collision Detection (CSMA/CD), Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA), Switched Local Area Networks

Transmission medium and physical layer



Classes of transmission media



GUIDED MEDIA:

Guided media, which are those that provide a channel from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.

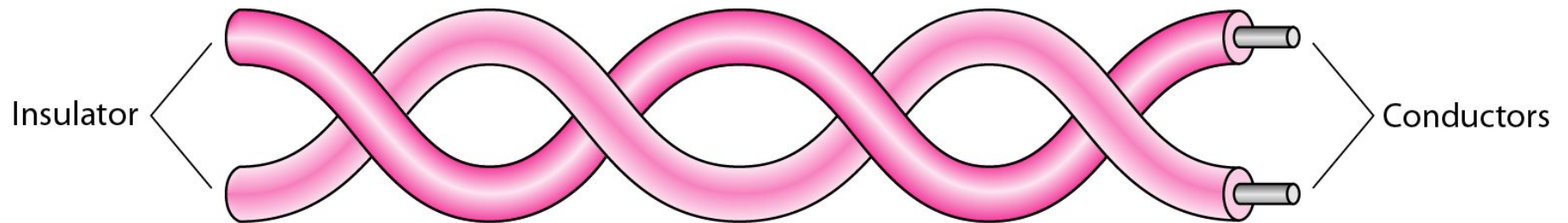
Topics discussed in this section:

Twisted-Pair Cable

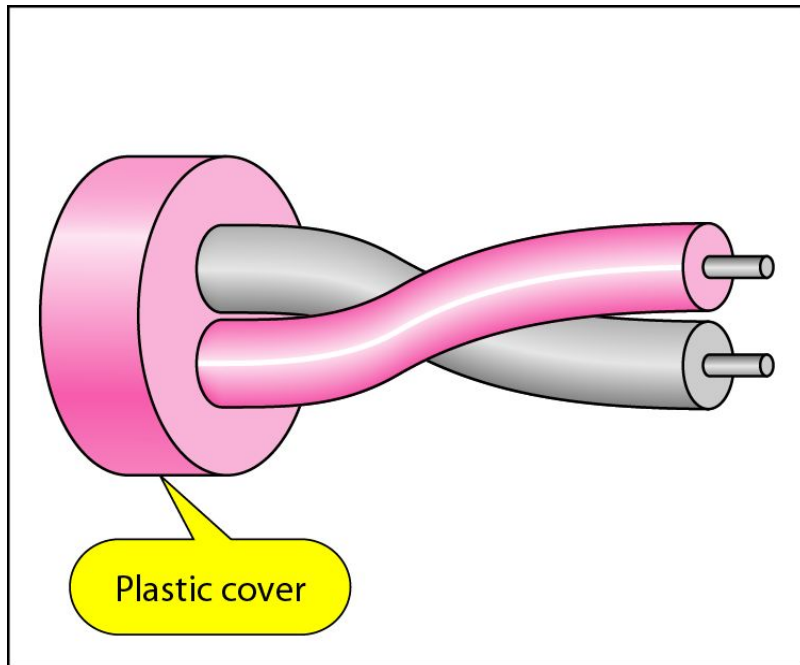
Coaxial Cable

Fiber-Optic Cable

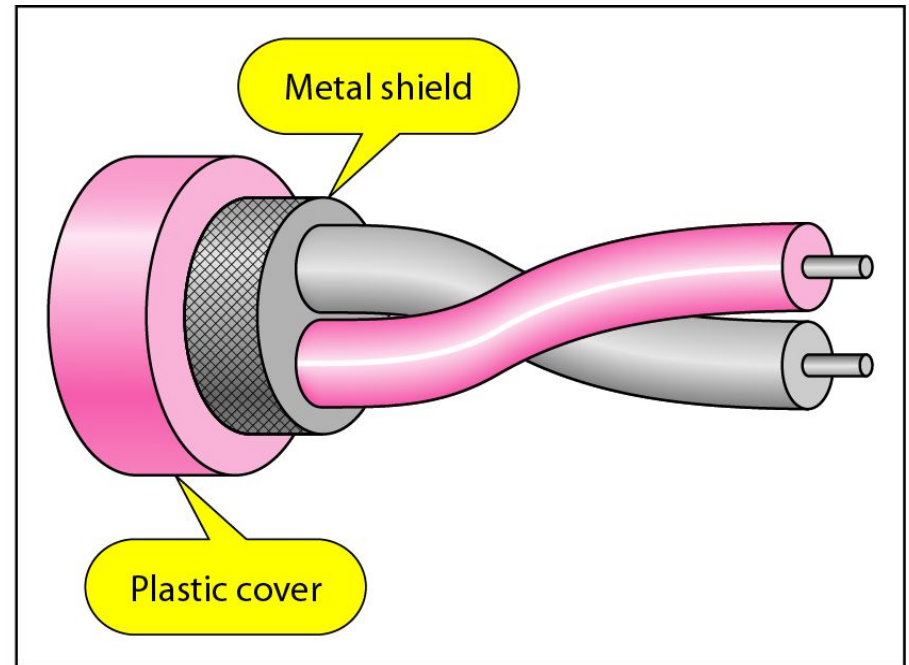
Twisted-pair cable



UTP and STP cables



a. UTP

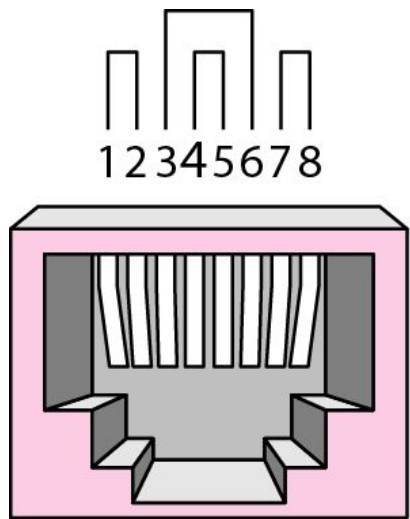


b. STP

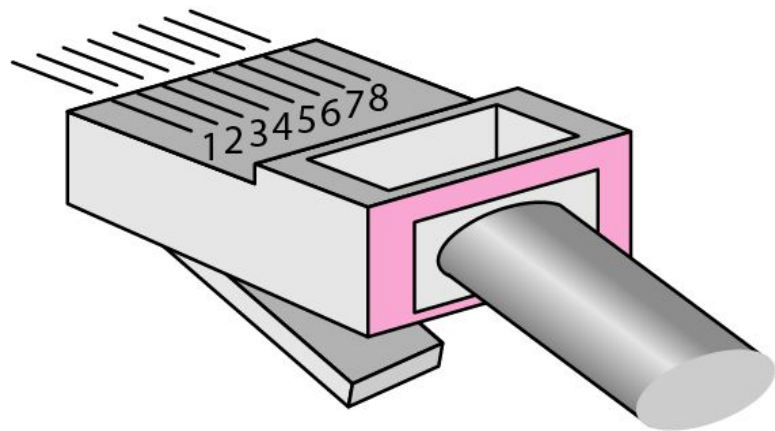
Categories of unshielded twisted-pair cables

<i>Category</i>	<i>Specification</i>	<i>Data Rate (Mbps)</i>	<i>Use</i>
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T-lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called SSTP (shielded screen twisted-pair). Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

UTP connector

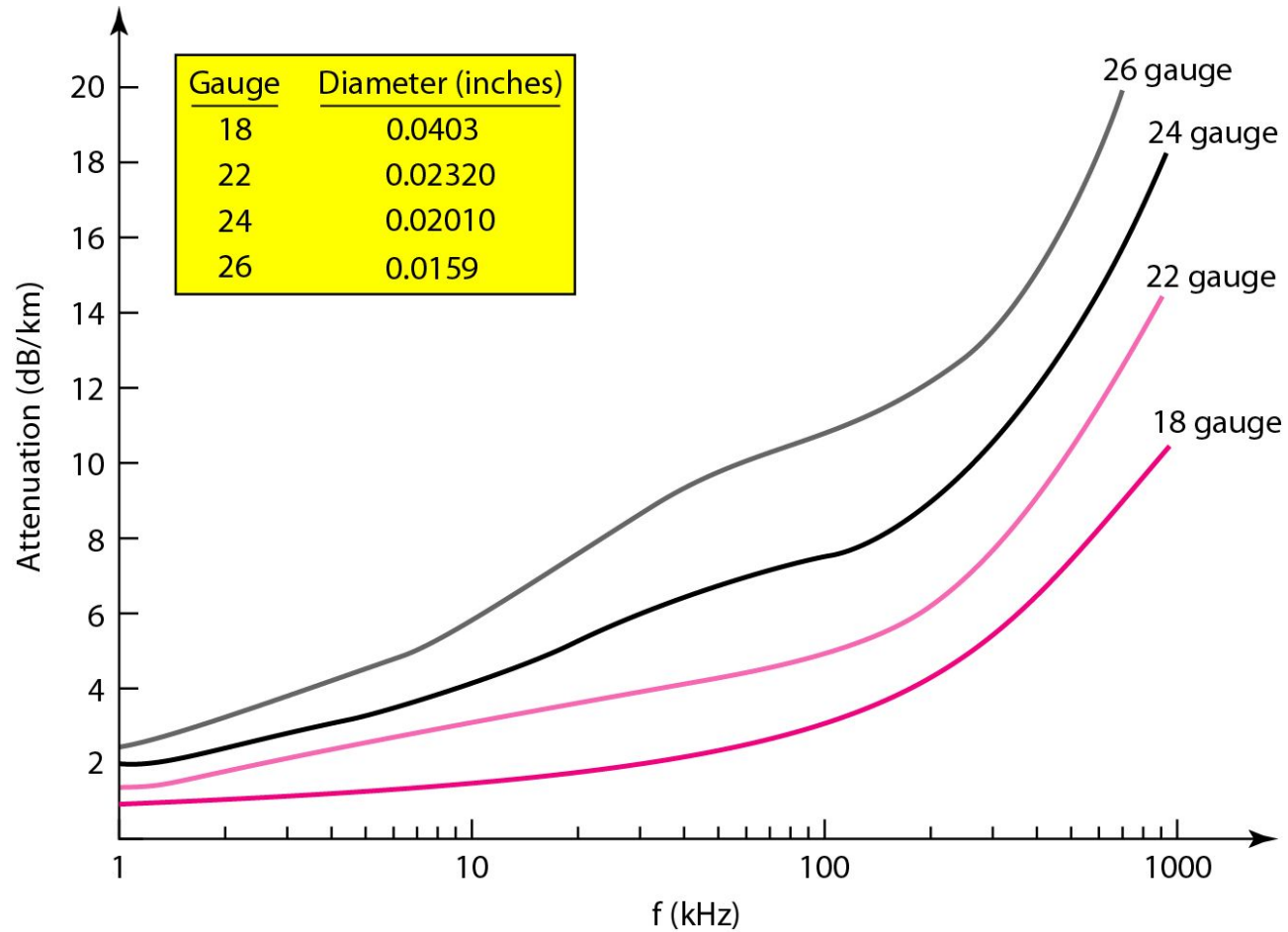


RJ-45 Female

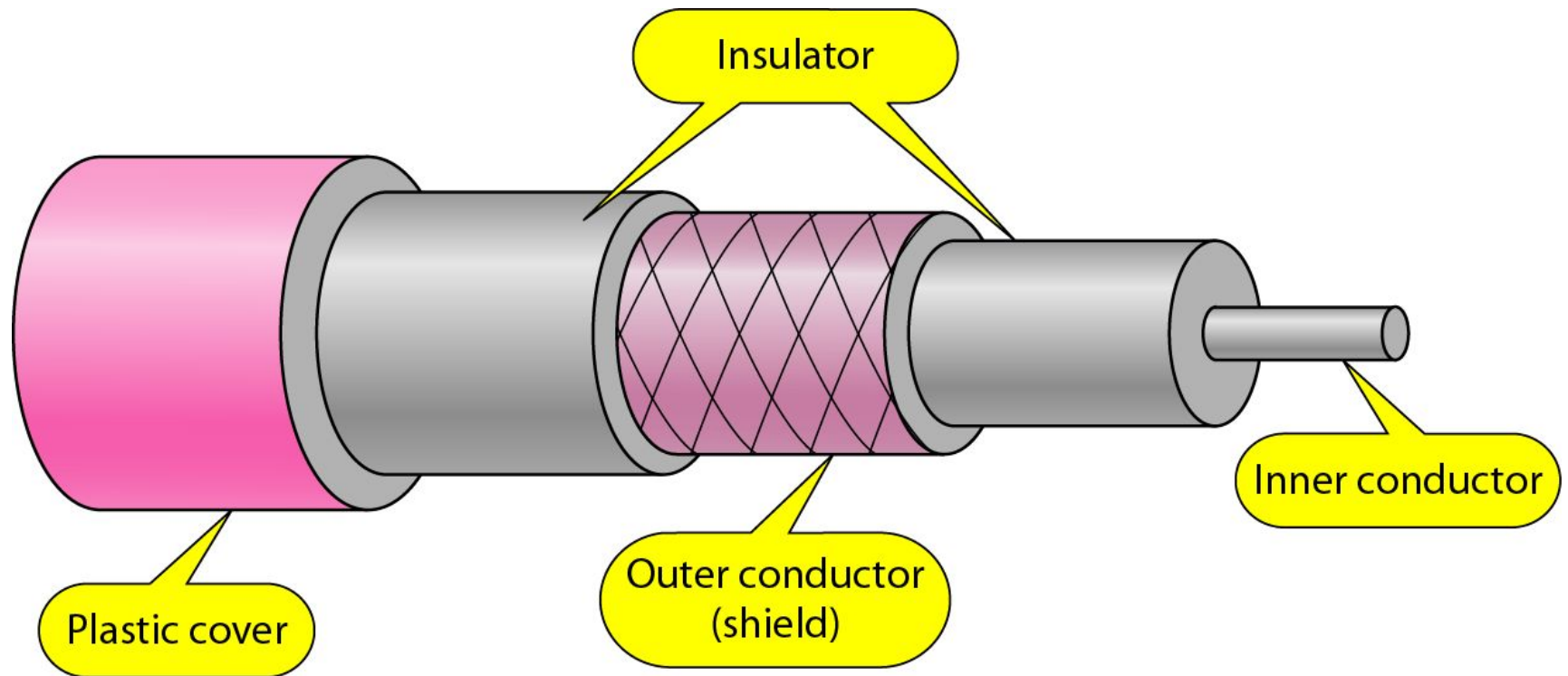


RJ-45 Male

UTP performance



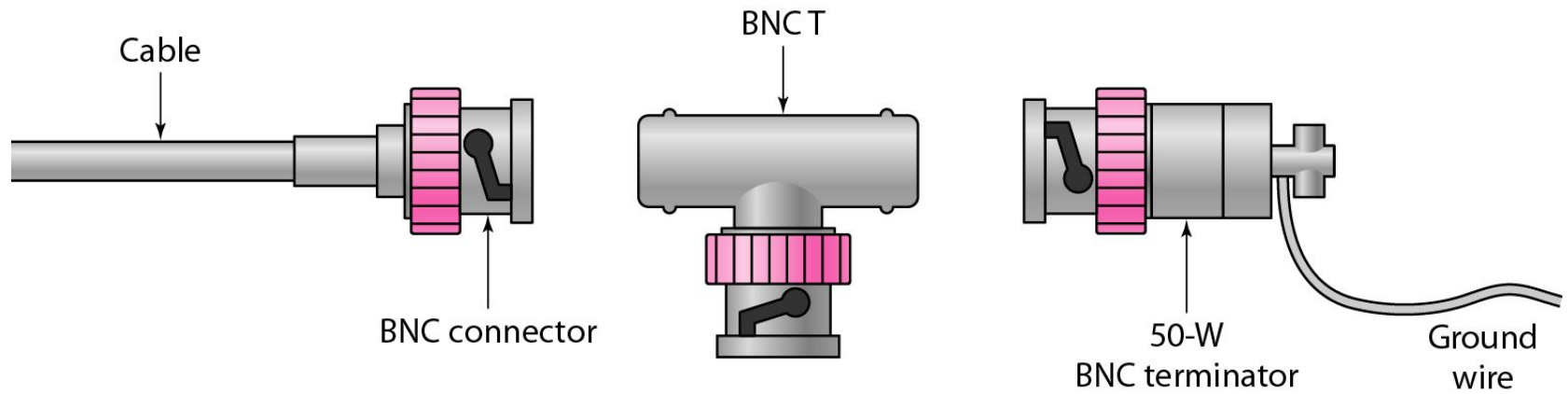
Coaxial cable



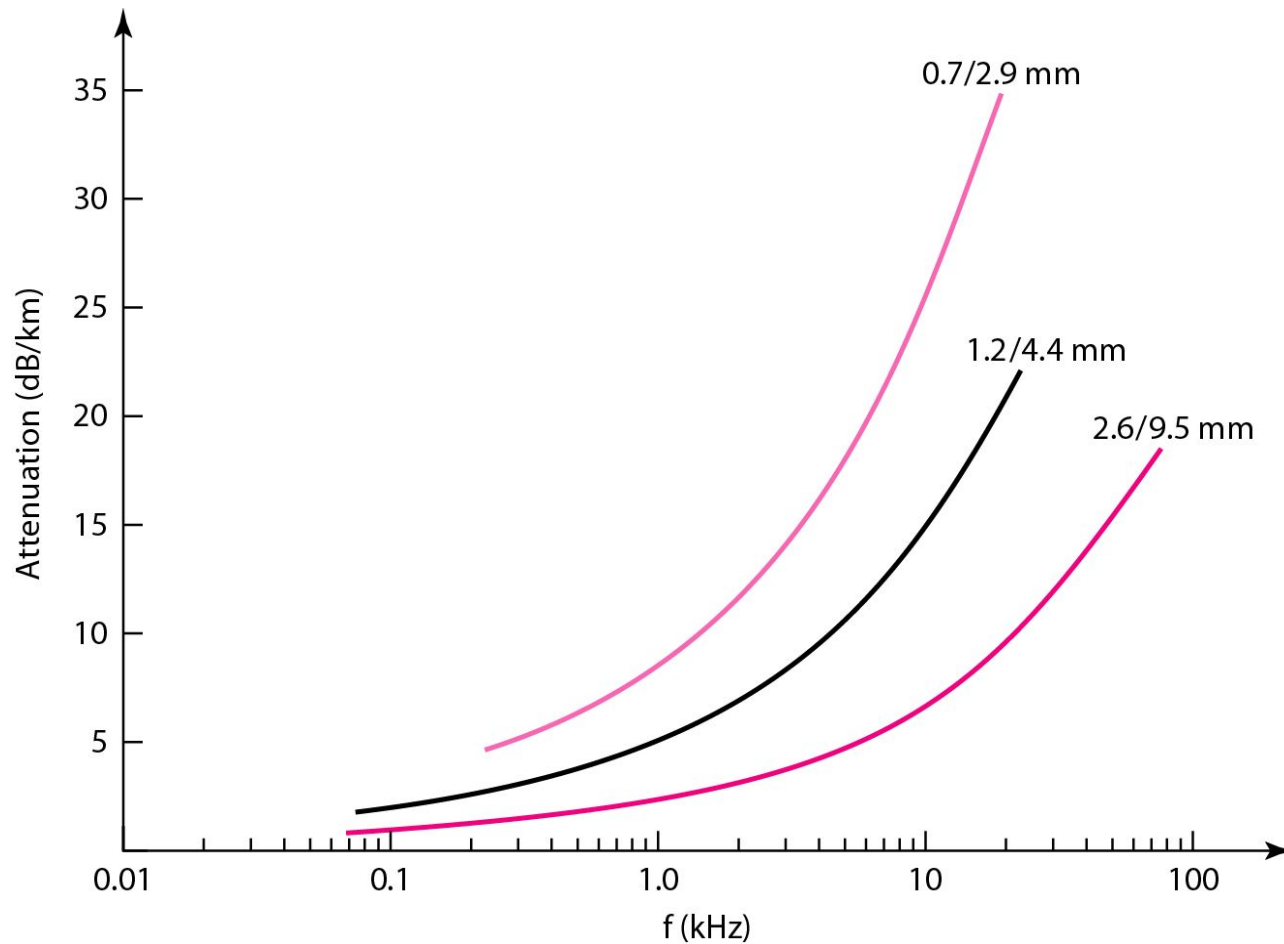
Categories of coaxial cables

<i>Category</i>	<i>Impedance</i>	<i>Use</i>
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

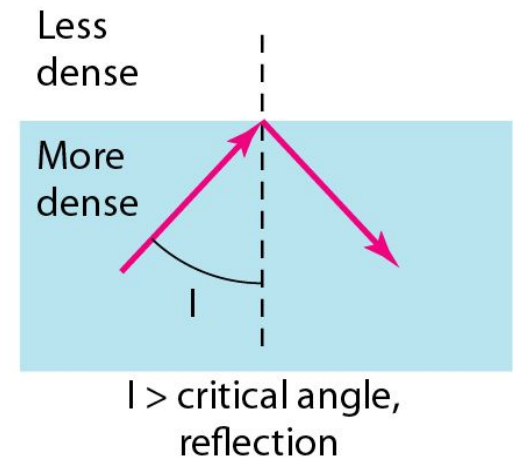
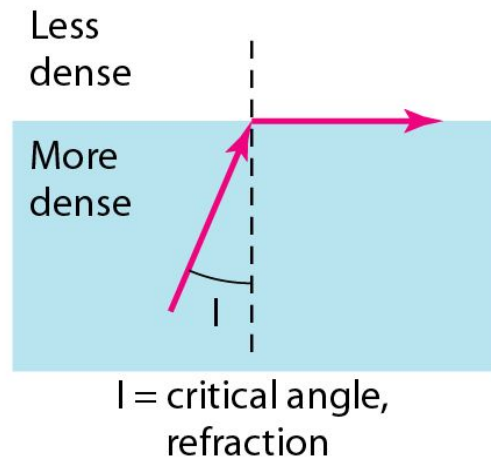
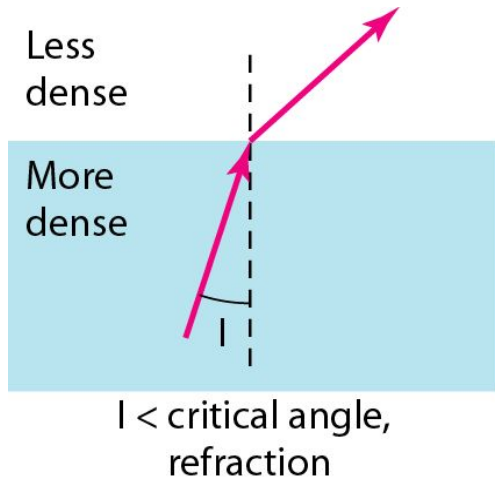
BNC connectors



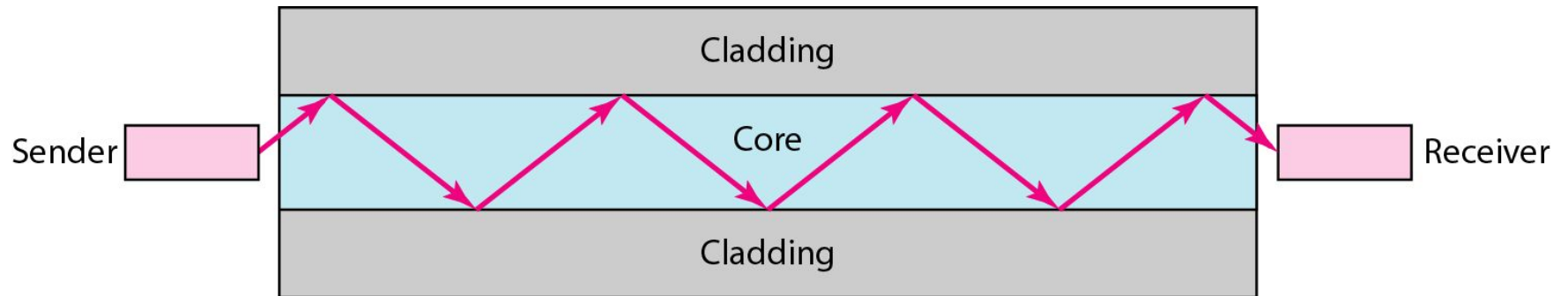
Coaxial cable performance



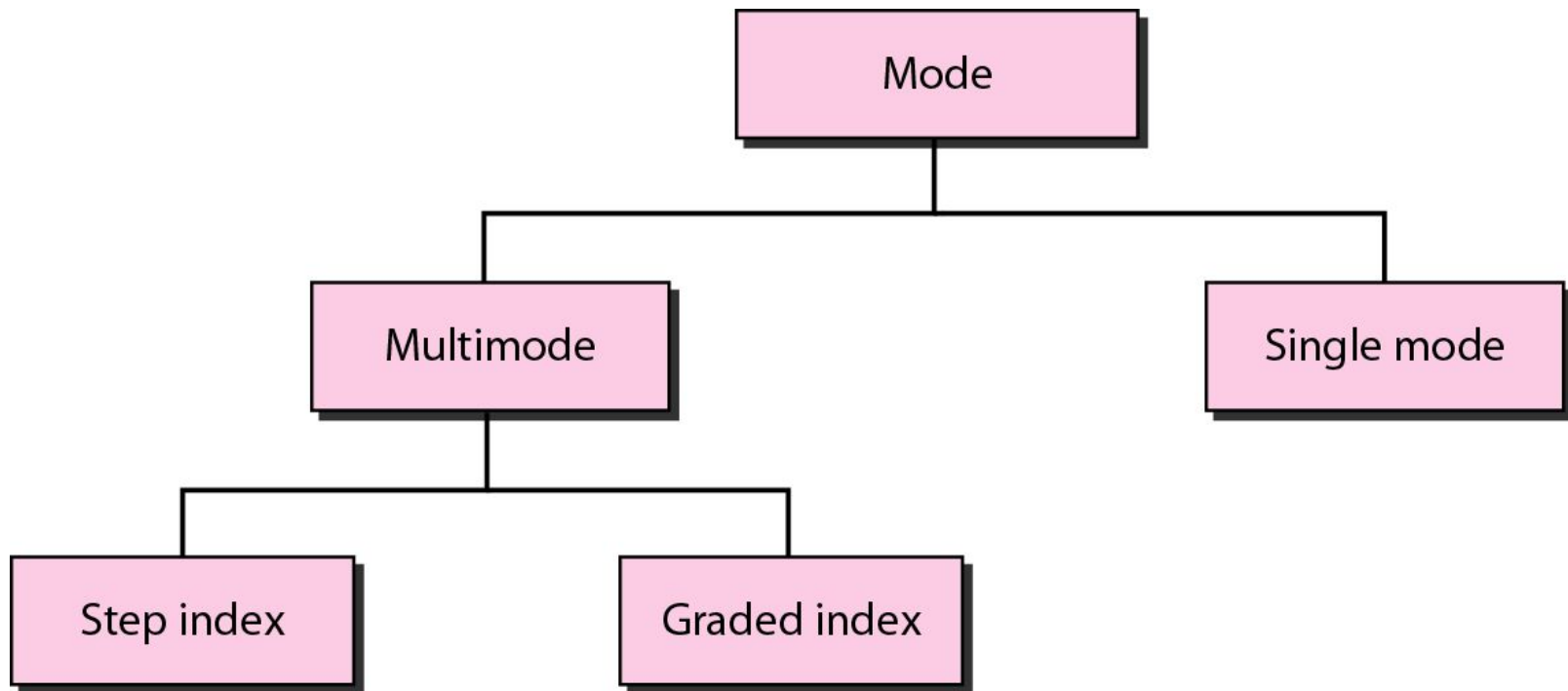
Fiber optics: *Bending of light ray*



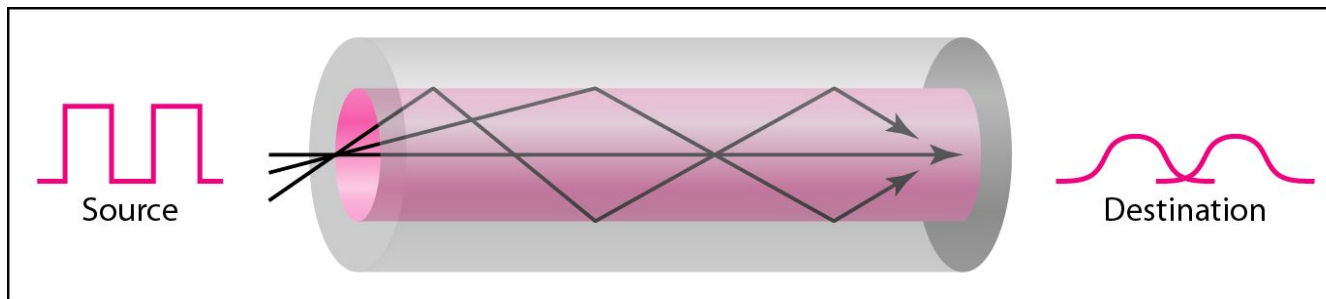
Optical fiber



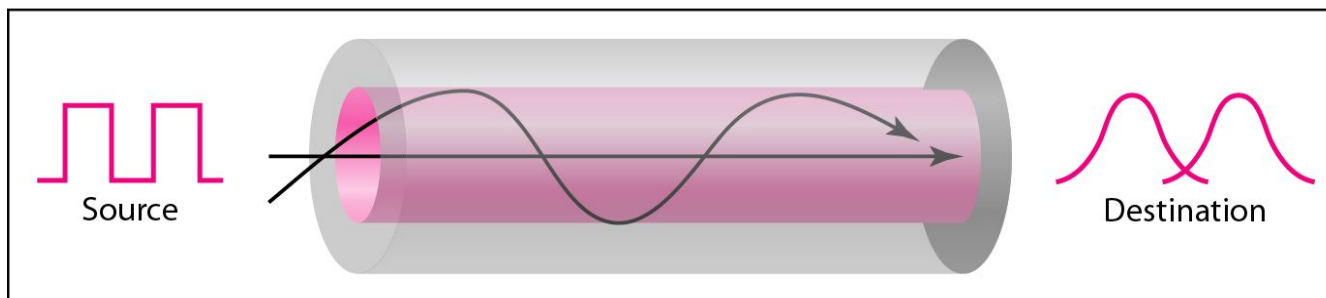
Propagation modes



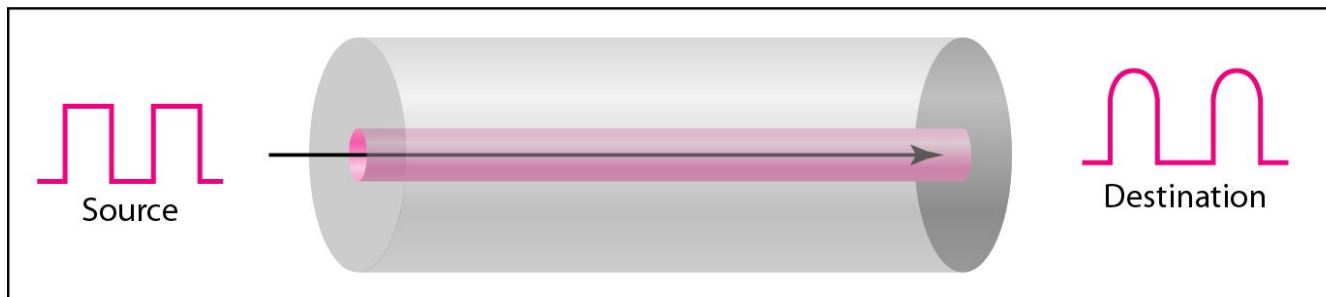
Modes



a. Multimode, step index



b. Multimode, graded index

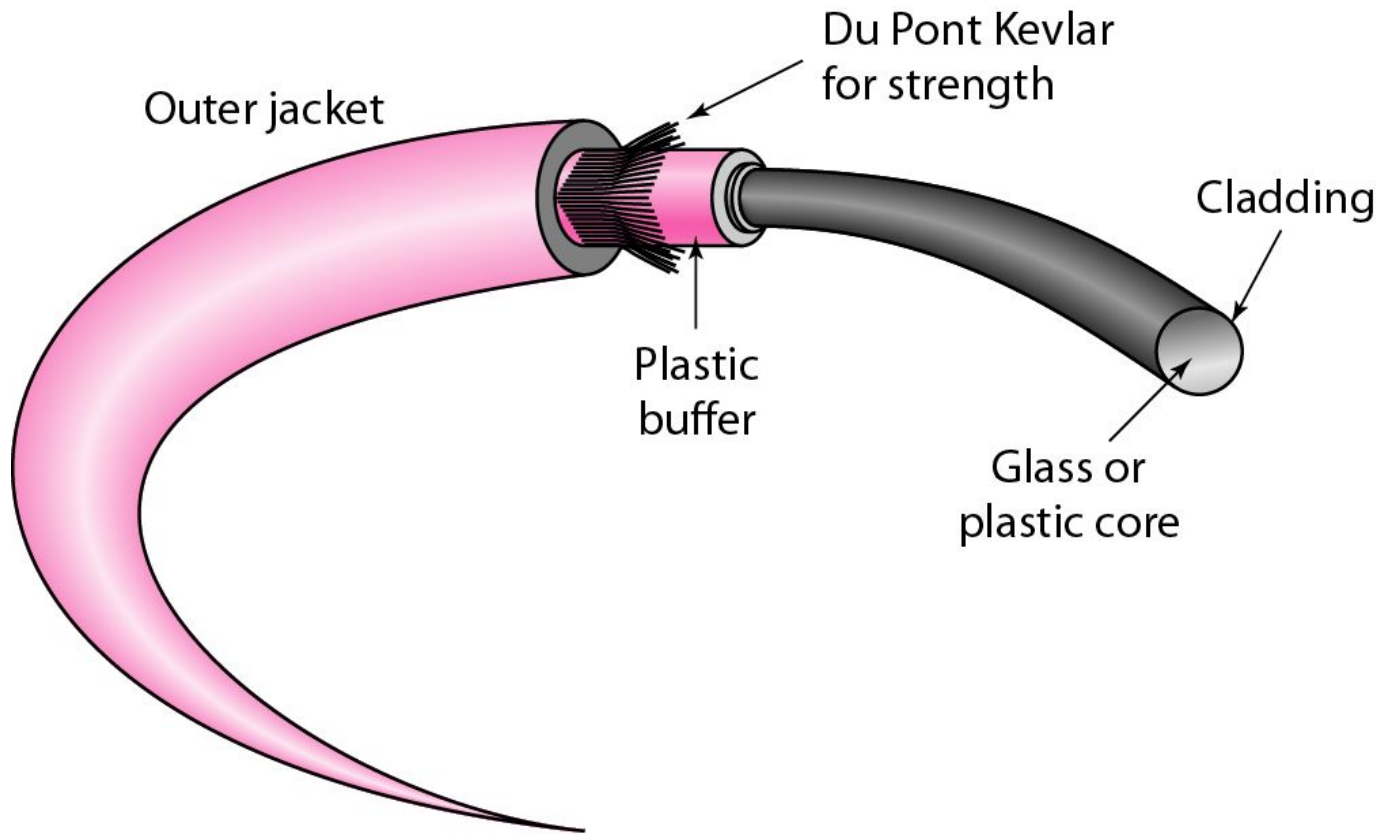


c. Single mode

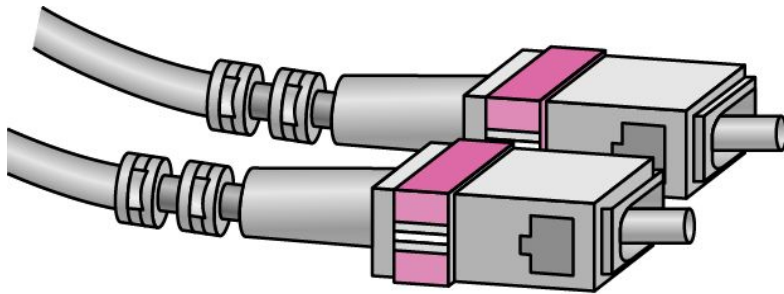
Fiber types

<i>Type</i>	<i>Core (μm)</i>	<i>Cladding (μm)</i>	<i>Mode</i>
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

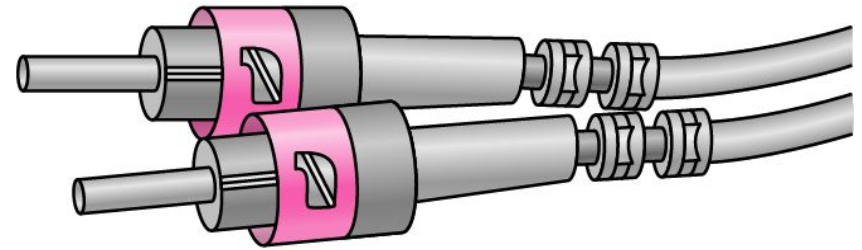
Fiber construction



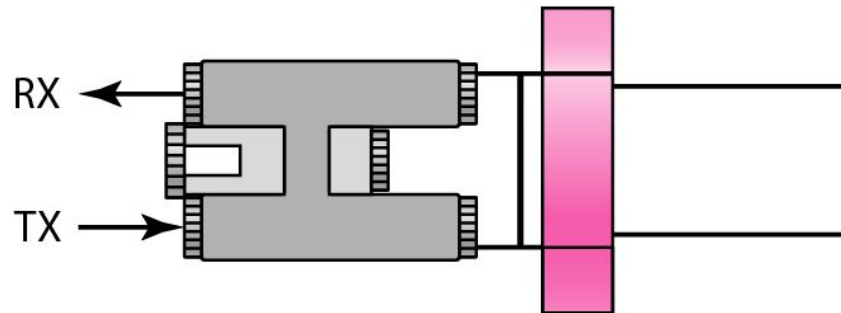
Fiber-optic cable connectors



SC connector
Subscriber channel

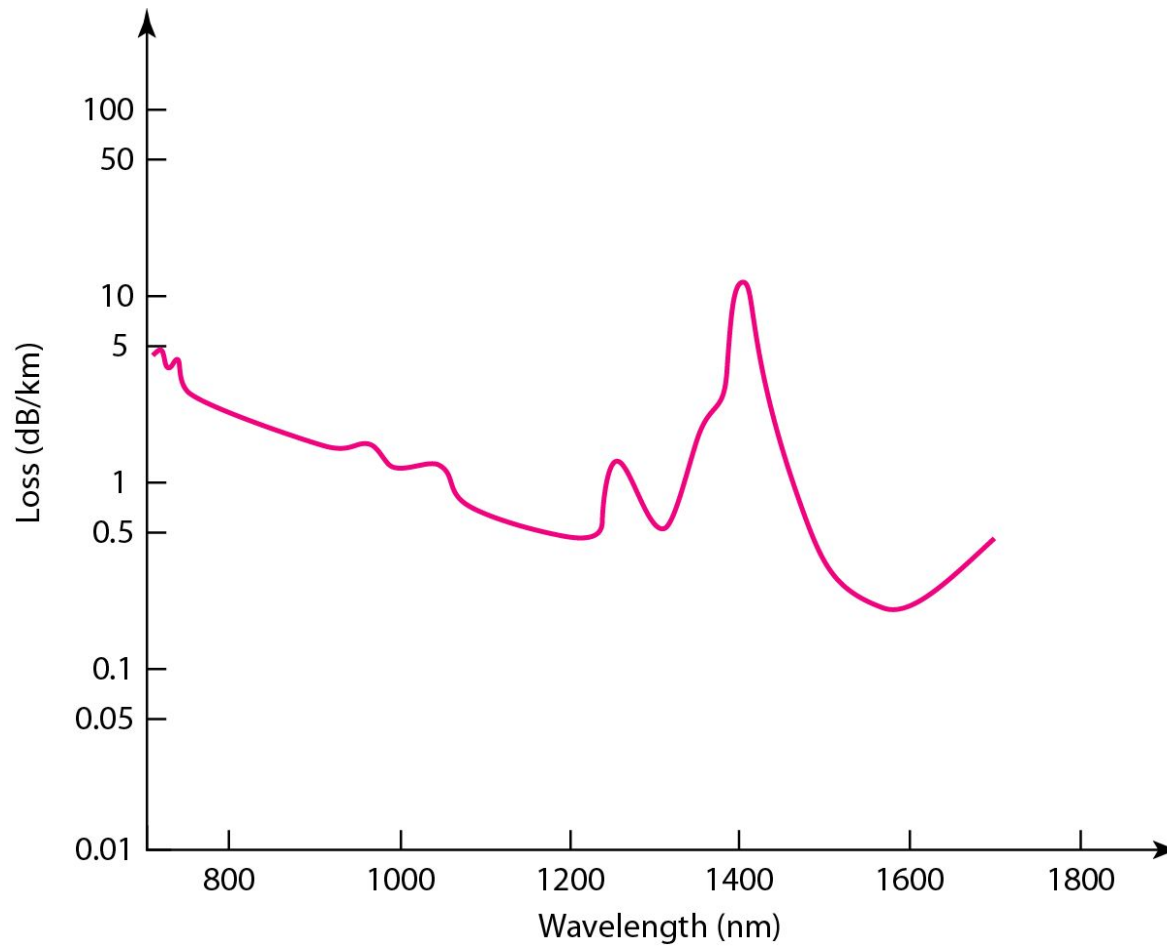


ST connector
straight-tip



MT-RJ connector

Optical fiber performance



UNGUIDED MEDIA: WIRELESS

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.

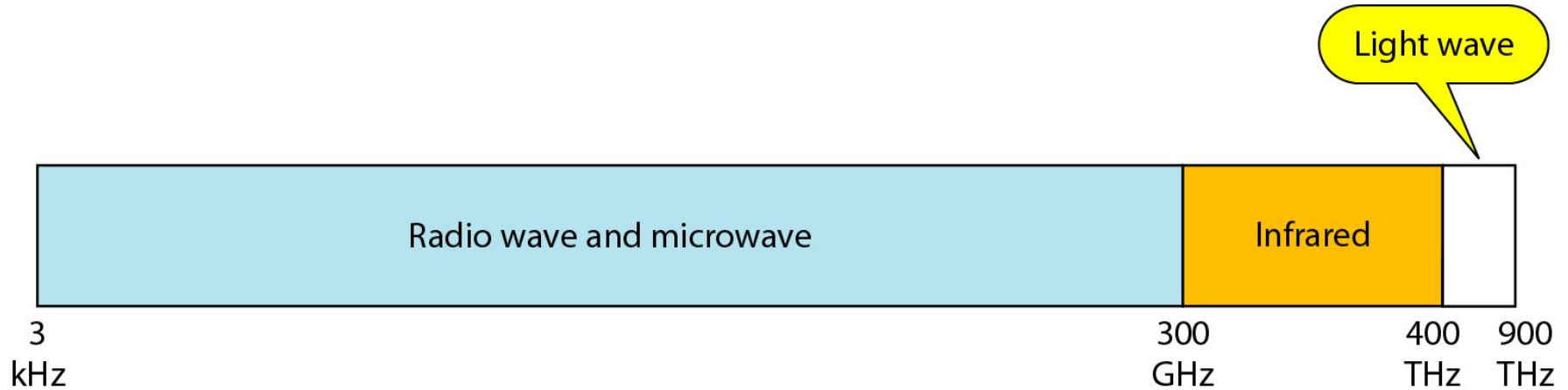
Topics discussed in this section:

Radio Waves

Microwaves

Infrared

Electromagnetic spectrum for wireless communication:



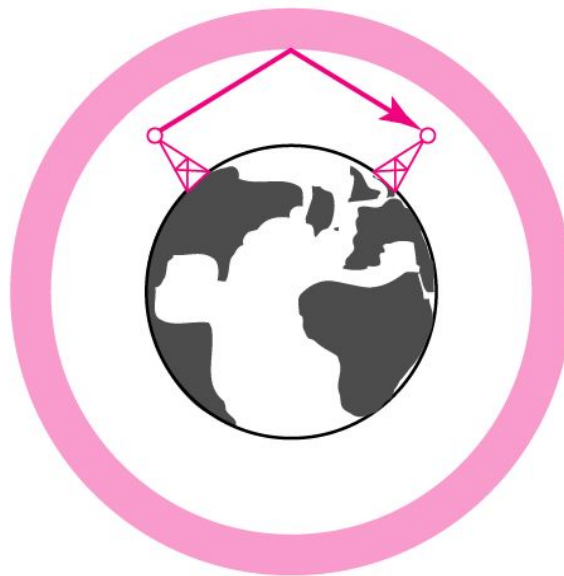
Propagation methods

Ionosphere



Ground propagation
(below 2 MHz)

Ionosphere



Sky propagation
(2–30 MHz)

Ionosphere

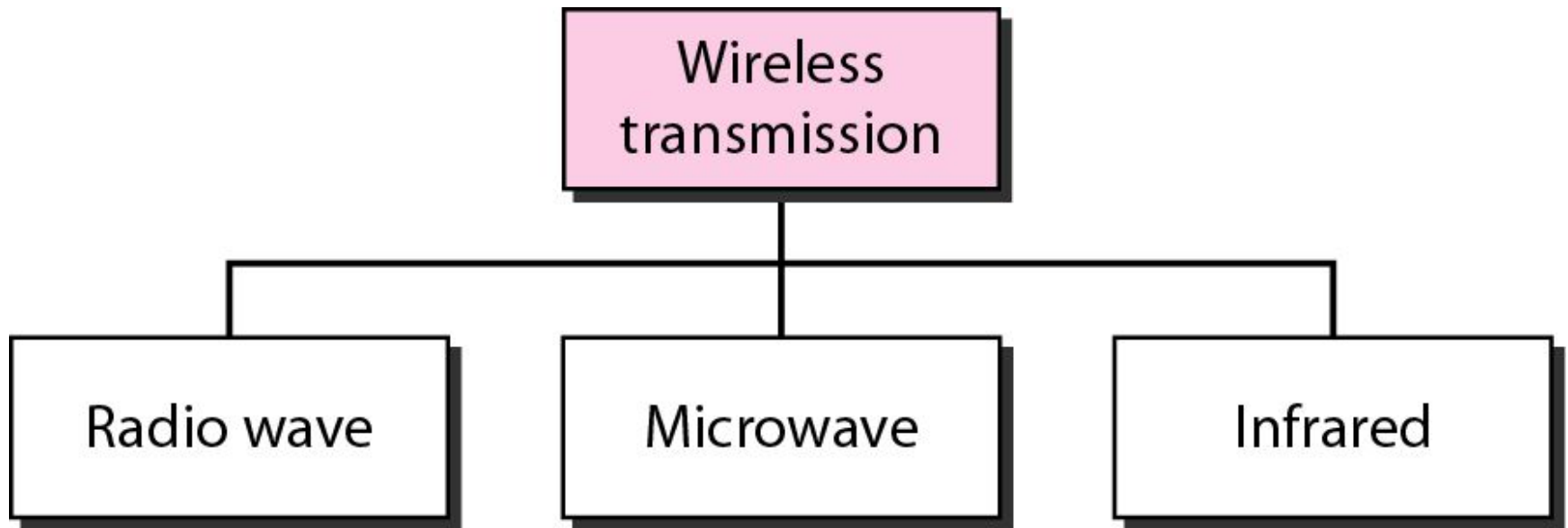


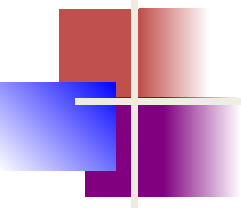
Line-of-sight propagation
(above 30 MHz)

Bands

<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

Wireless transmission waves

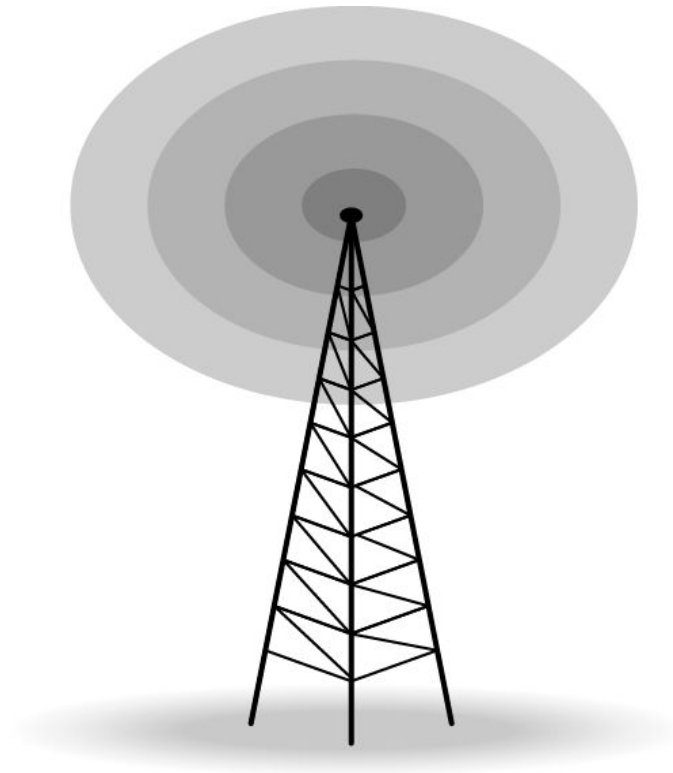


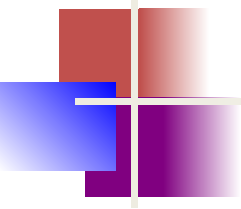


Radio Waves

- ❑ Radio waves are used for multicast communications, such as radio and television, and paging systems.
 - ❑ They can penetrate through walls.
 - ❑ Highly regulated.
 - ❑ Use Omni directional antennas
-

Omnidirectional antenna

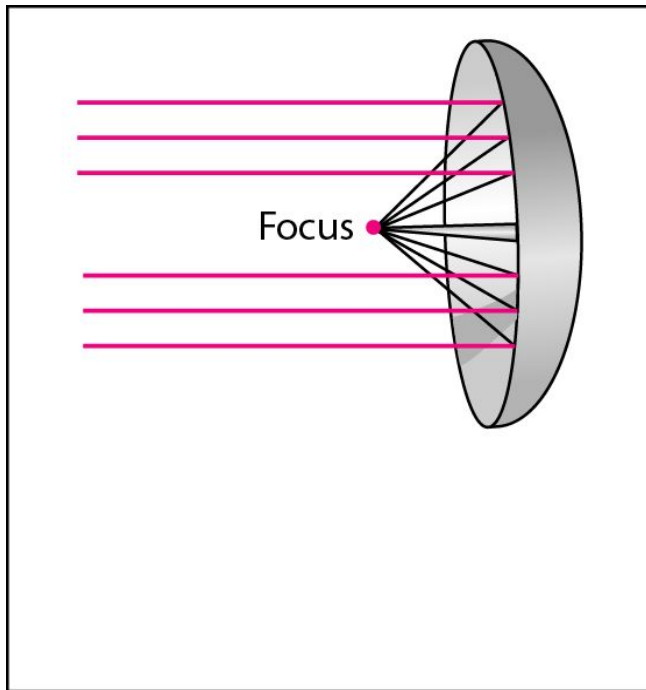




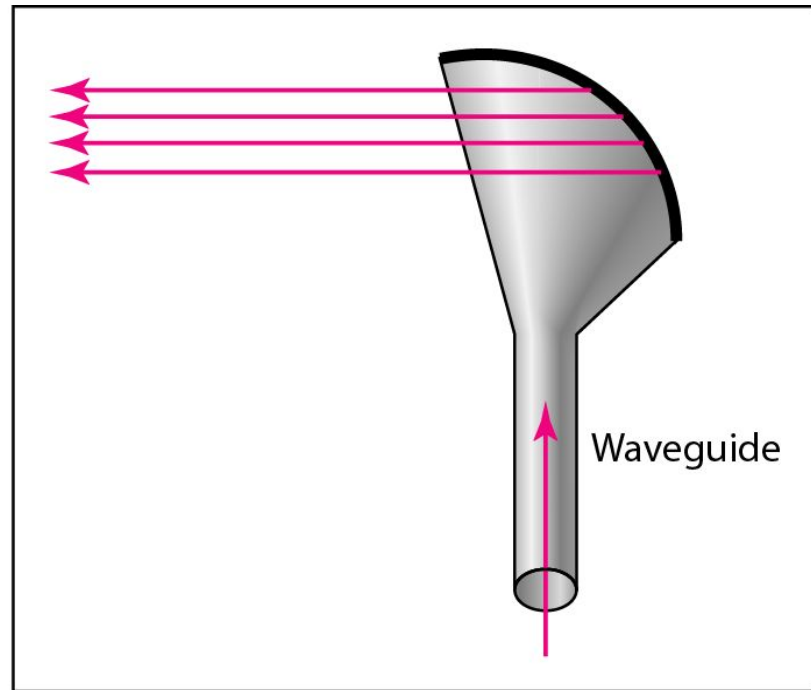
Microwaves

- Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.
 - Higher frequency ranges cannot penetrate walls.
 - Use directional antennas - point to point line of sight communications.
-

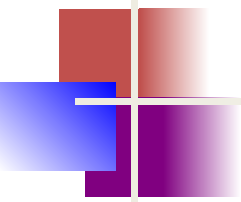
Unidirectional antennas



a. Dish antenna



b. Horn antenna



Infrared

Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

Wireless Channels:

- Are subject to a lot more errors than guided media channels.
- Interference is one cause for errors, can be circumvented with high SNR.
- The higher the SNR the less capacity is available for transmission due to the broadcast nature of the channel.
- Channel also subject to fading and no coverage holes.

Switching

Outline

- Switched Network
- Circuit-Switched Network
- Datagram Networks
- Virtual-Circuit Network



Introduction

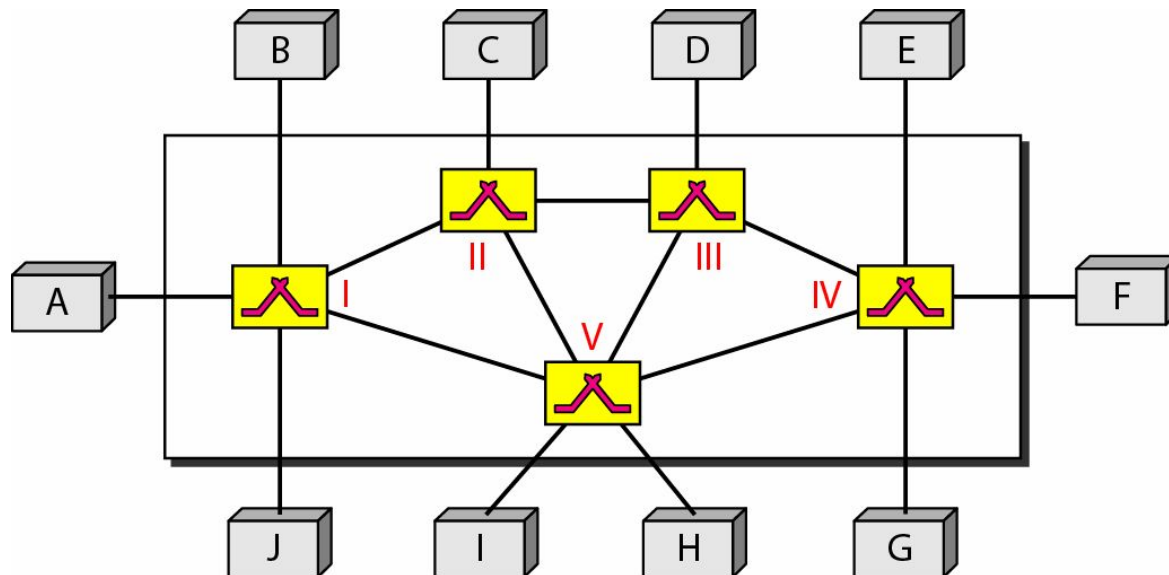
- In large networks we need some means to allow one-to-one communication between any two nodes.
- In LANs this is achieved using one of three methods:
 - Direct point-to-point connection (mesh)
 - Via central controller (star)
 - Connection to common bus in a multipoint configuration (bus/hup)

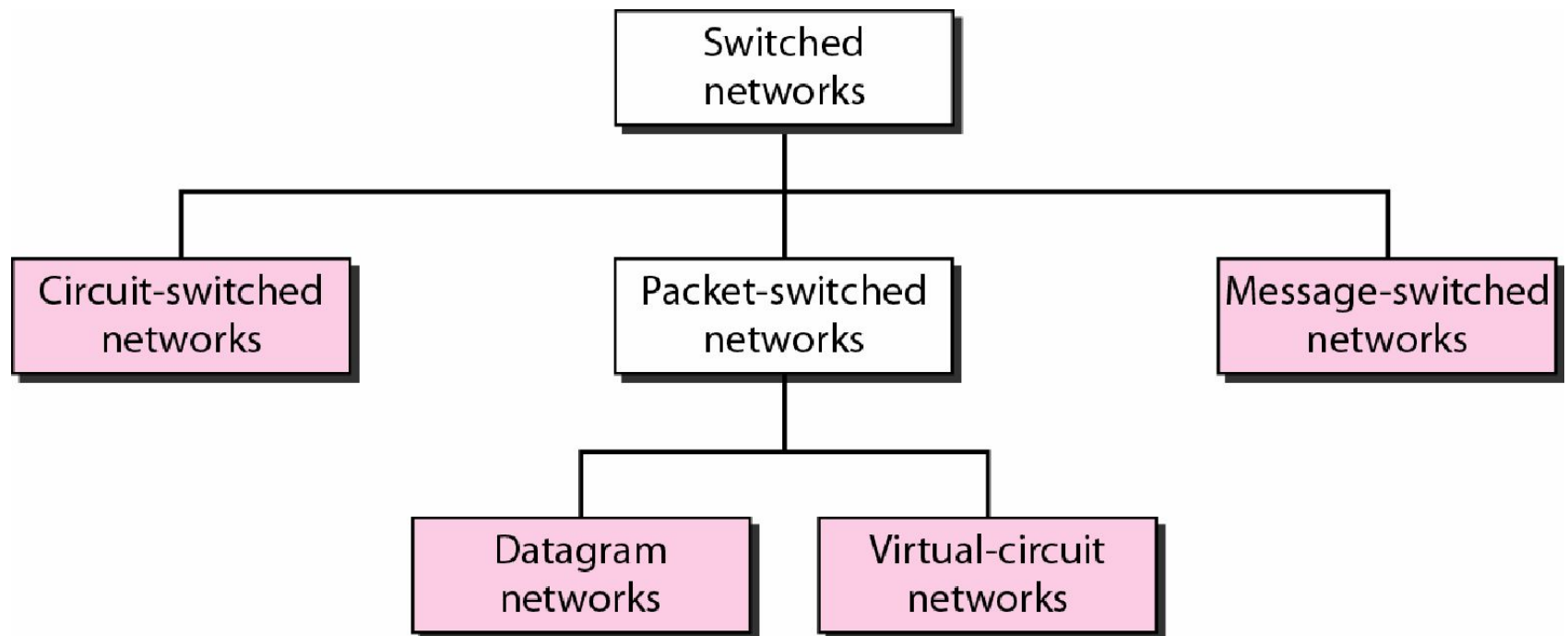
Introduction

- None of the previous works in larger networks with large physical separation or consisting of a large number of computers
- The solution is a switching network

Switched Network

- Consists of a series of interlinked nodes called switched.
- Switches are capable to create temporary connections between two or more devices

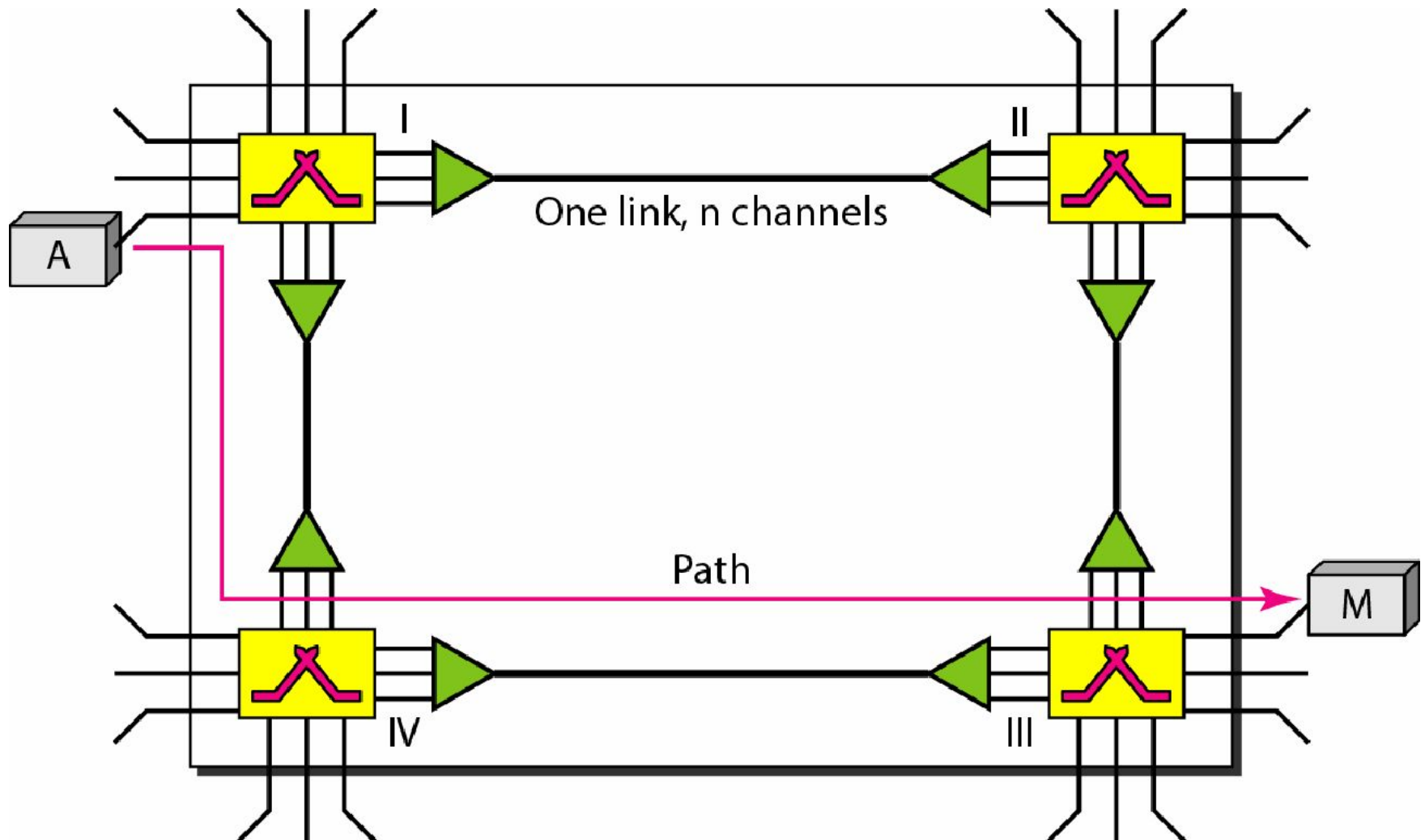




Circuit-Switched Network

- A circuit-switched network consists of a set of switches connected by physical links.
- A connection between two stations is a dedicated path made of one or more links
- each connection uses only one dedicated channel on each link
- Each link is normally divided into n channels by using FDM or TDM.
- The link can be permanent (leased line) or temporary (telephone)

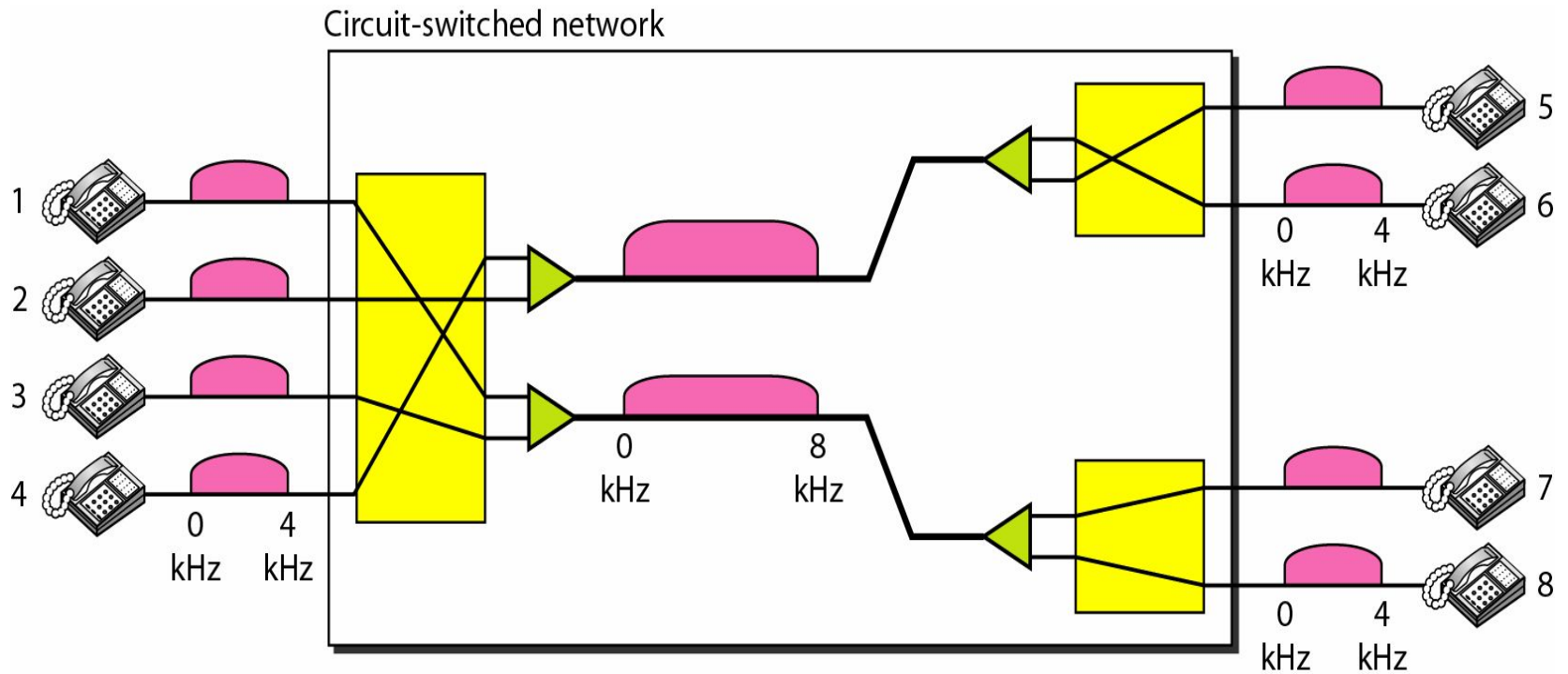
Circuit-Switched Network



Circuit-Switched Network

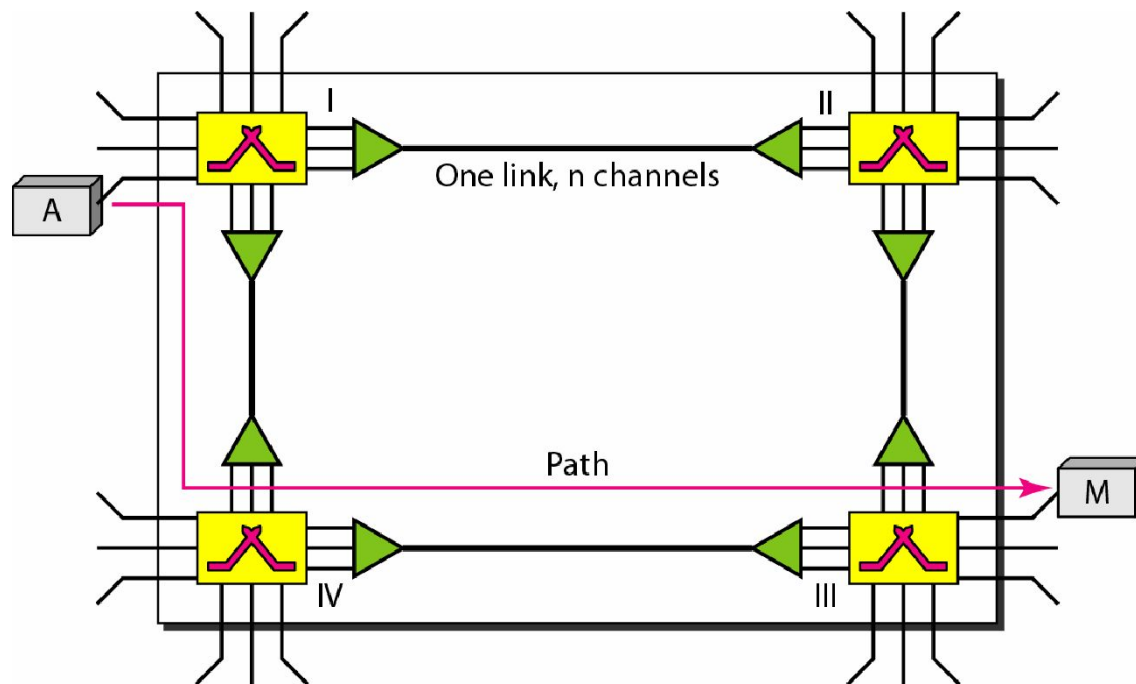
- Switching take place at physical layer
- Resources
 - Such as bandwidth in FDM and time slot in TDM
 - Switch buffer
 - Switch processing time
 - Switch I/O ports
- Data transferred are not packetized, continuous flow
- No addressing involved during data transfer

Example



Transmission phases

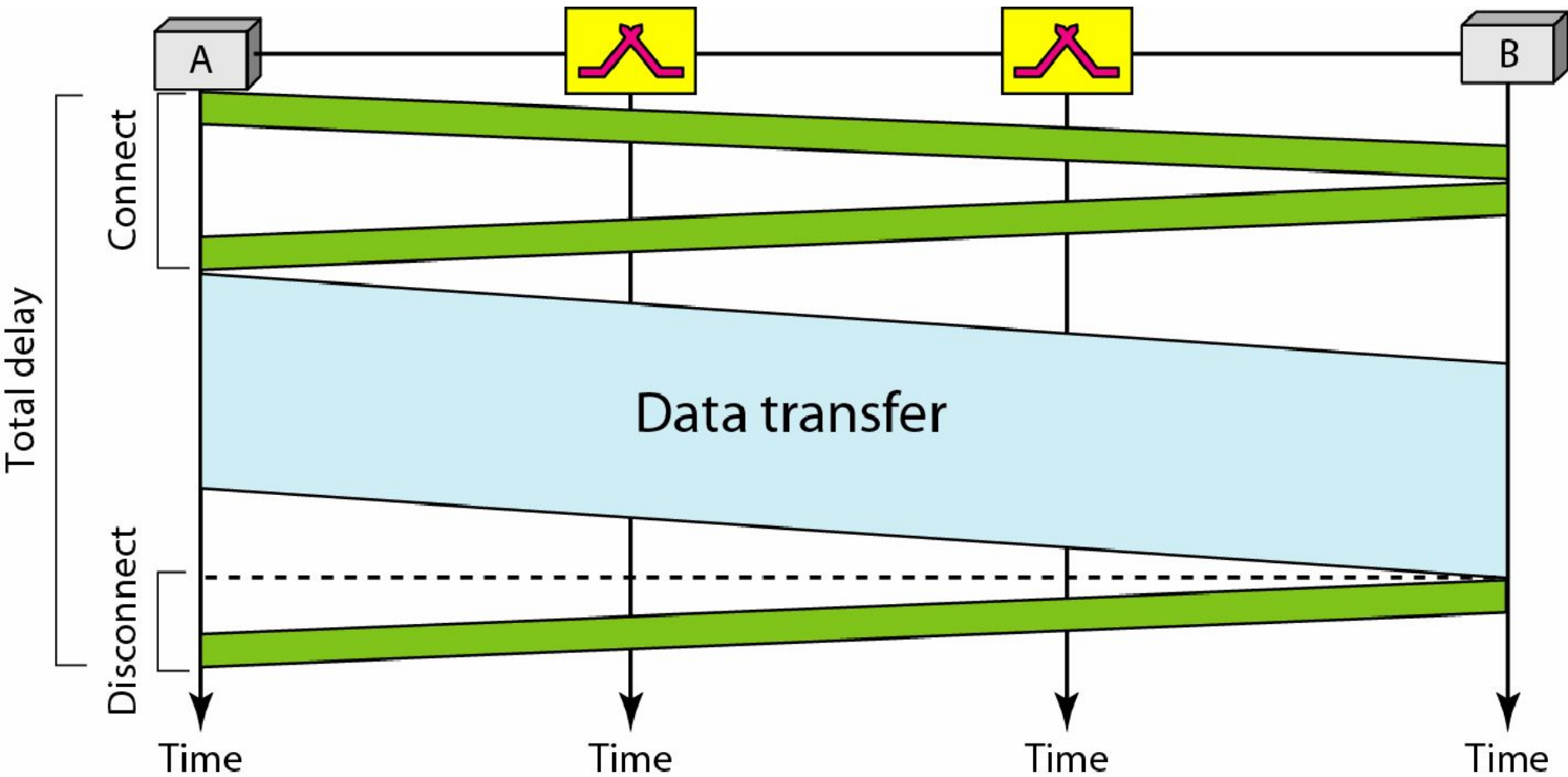
- Setup phase
 - A dedicated circuit needs to be established
 - So create dedicated channel by sending a request



Transmission phases

- Data transfer phase
- Teardown phase
 - Signal is sent to each switch to release resources

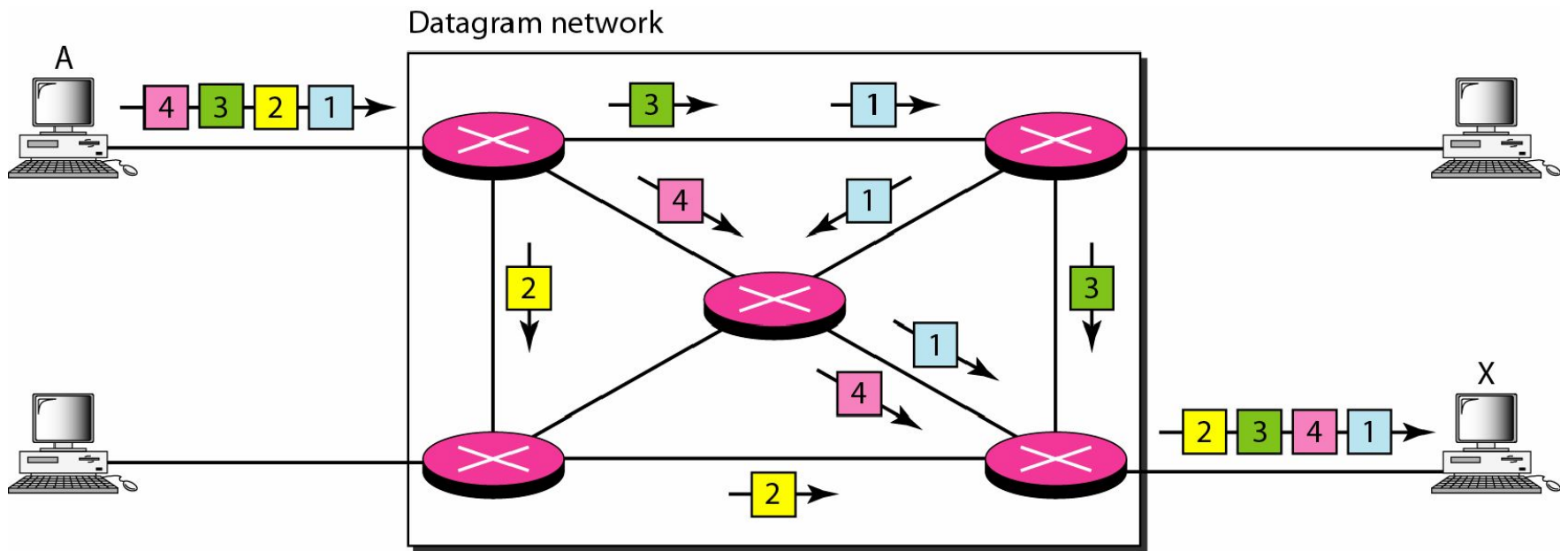
Delay



Datagram Networks

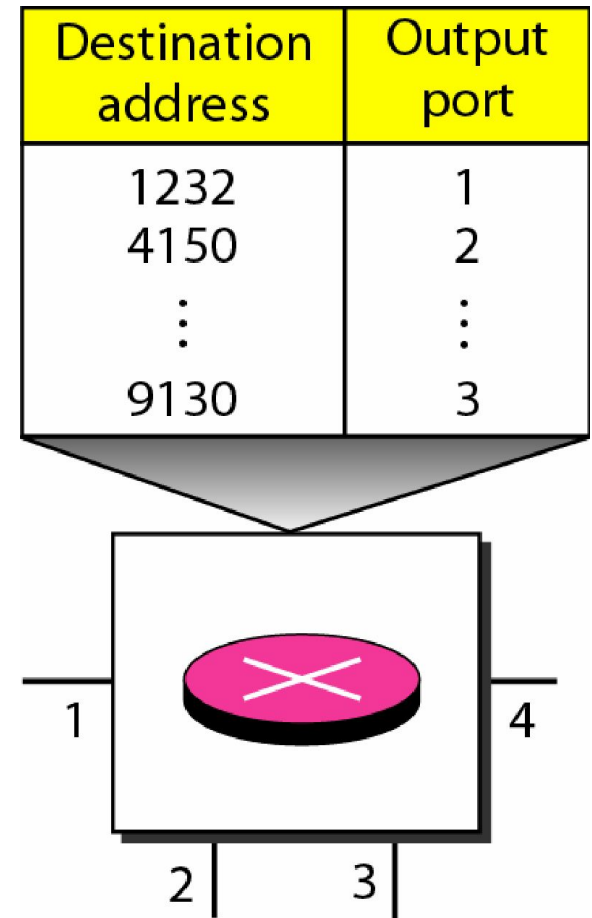
- Data are transmitted in discrete units called packets
- Size of the packet depends on the protocol and network
- Packets switched networks are connectionless, hence no resource allocation
- Connectionless means the switch does not keep information about the connection state.
- Datagram switching is done at network layer

Datagram Networks



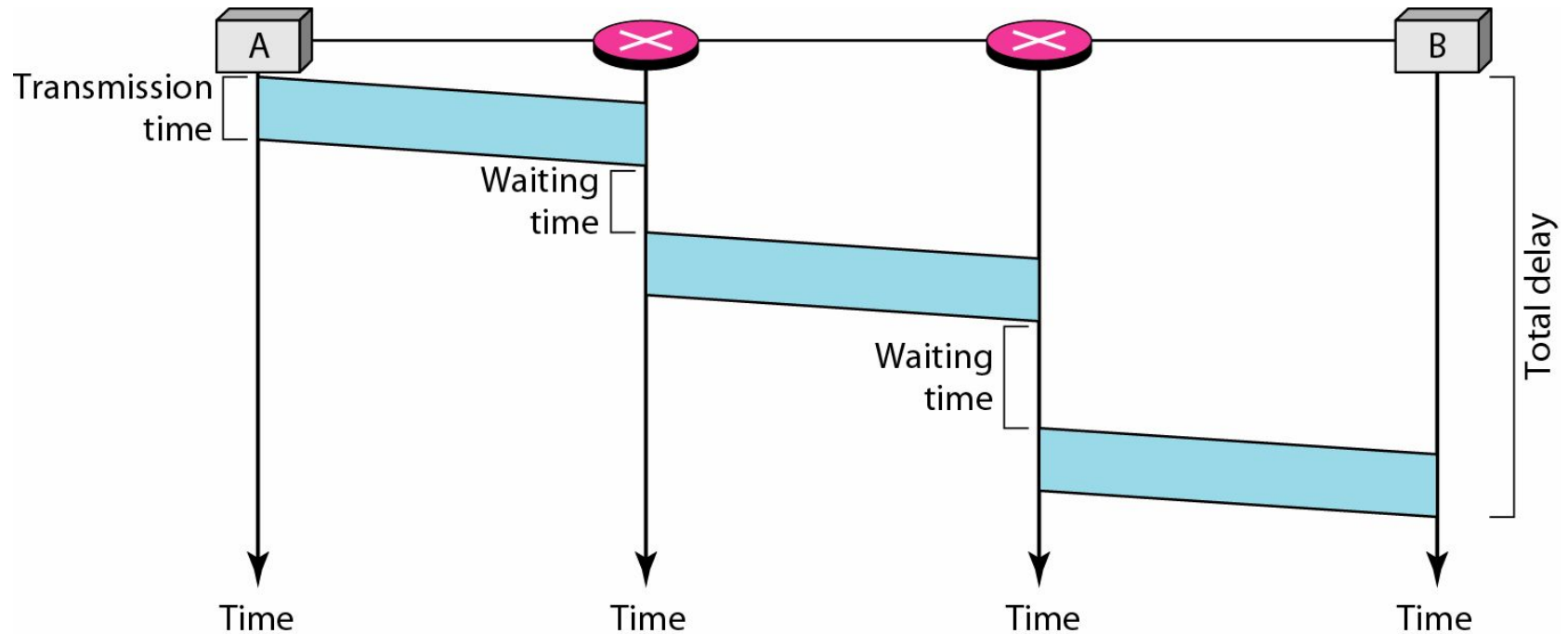
Routing table & Destination Add.

- A switch in a datagram network uses a routing table that is based on the destination address.
- The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet.



Delay

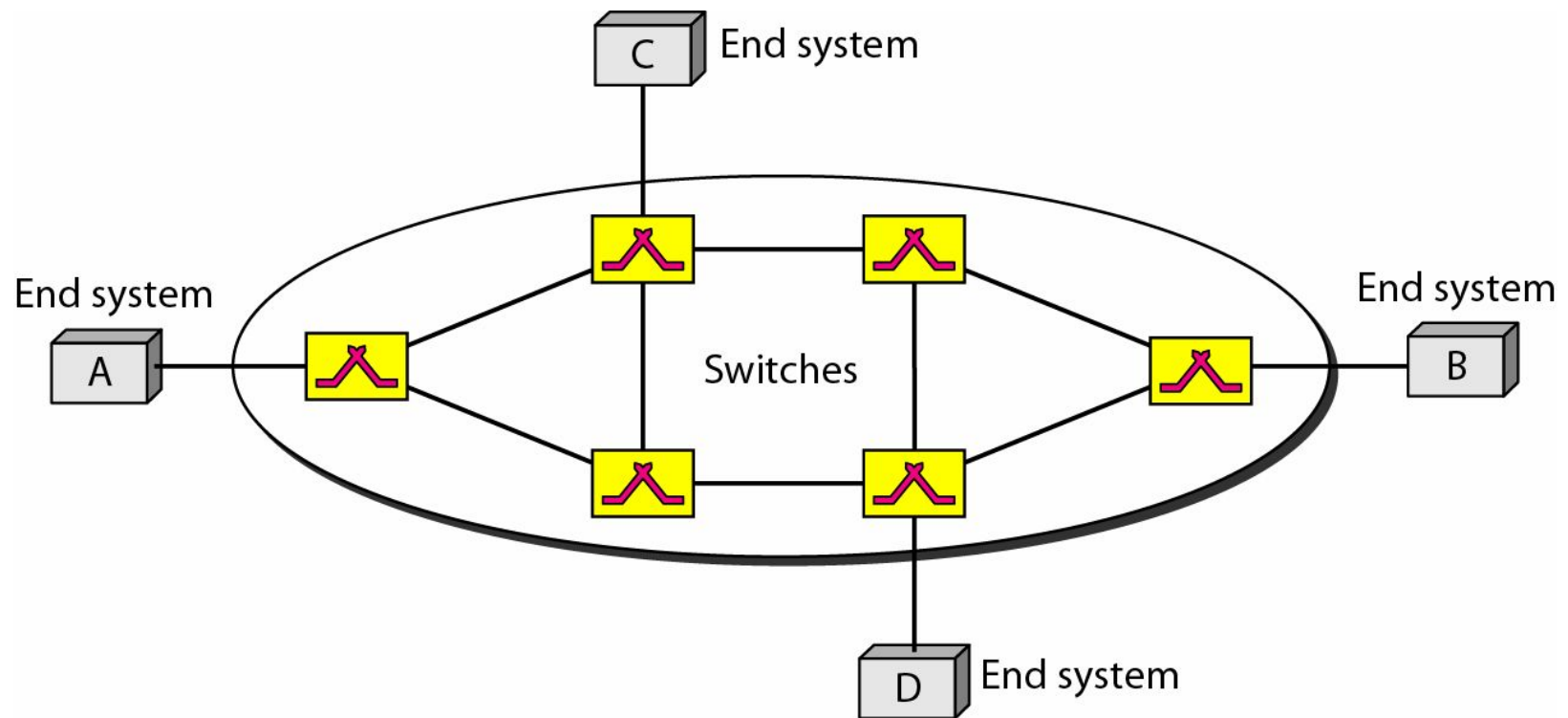
- Total Delay = $3T + 3t + w_1 + w_2$



Virtual-Circuit Network

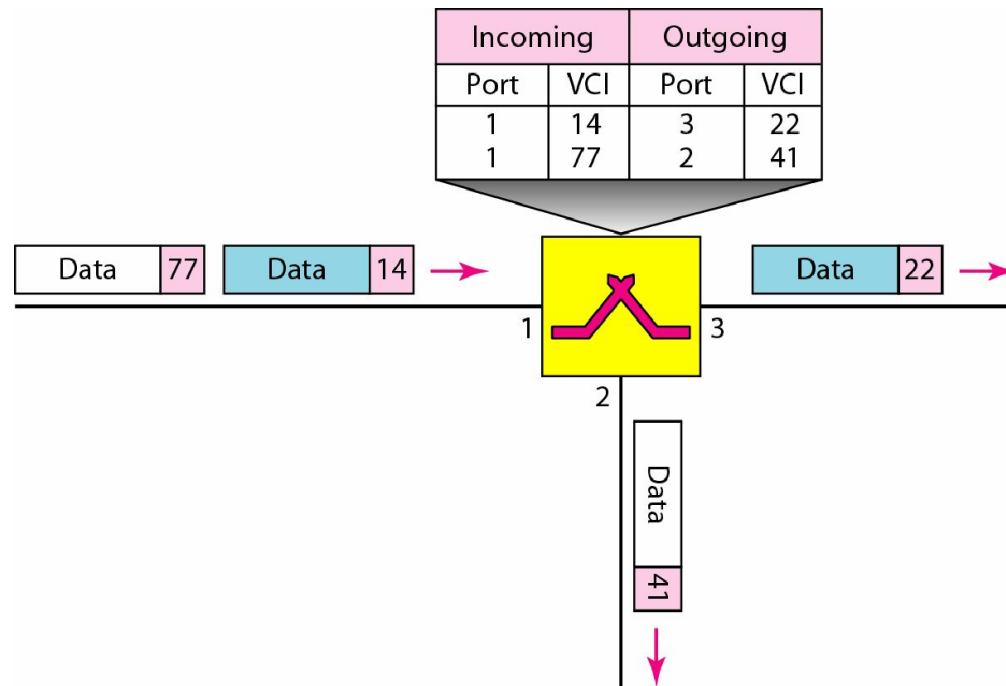
- Packets form a single message travel along the same path.
- Characteristics
 - Three phases to transfer data
 - Resources can be allocated during setup phase
 - Data are packetized and each packet carries an address in the header
 - All packets follow the same path
 - Implemented in data link layer

Virtual-Circuit Network



Addressing

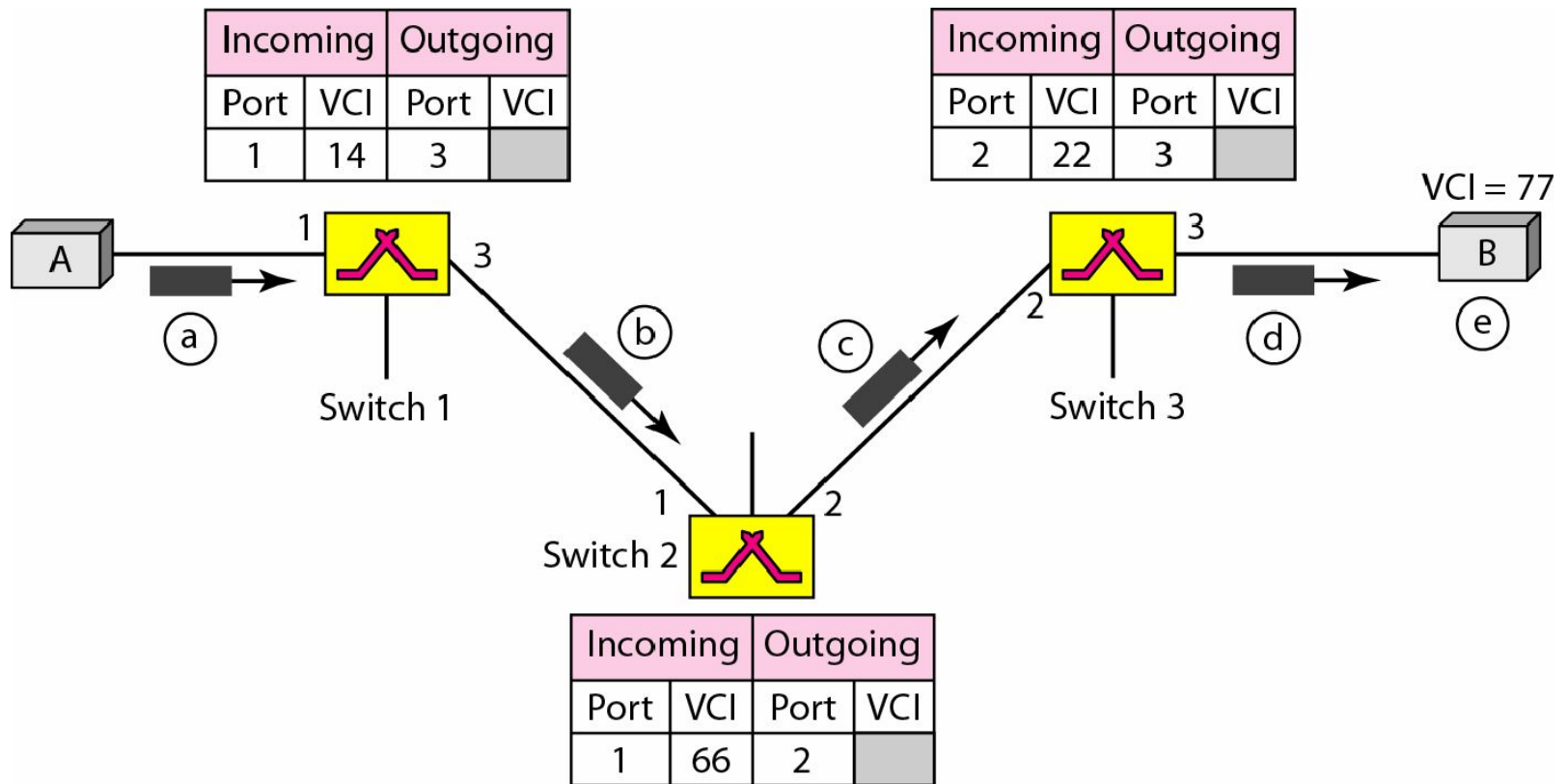
- Global addressing
 - Source and destination needs unique addresses
- Virtual-circuit identifier



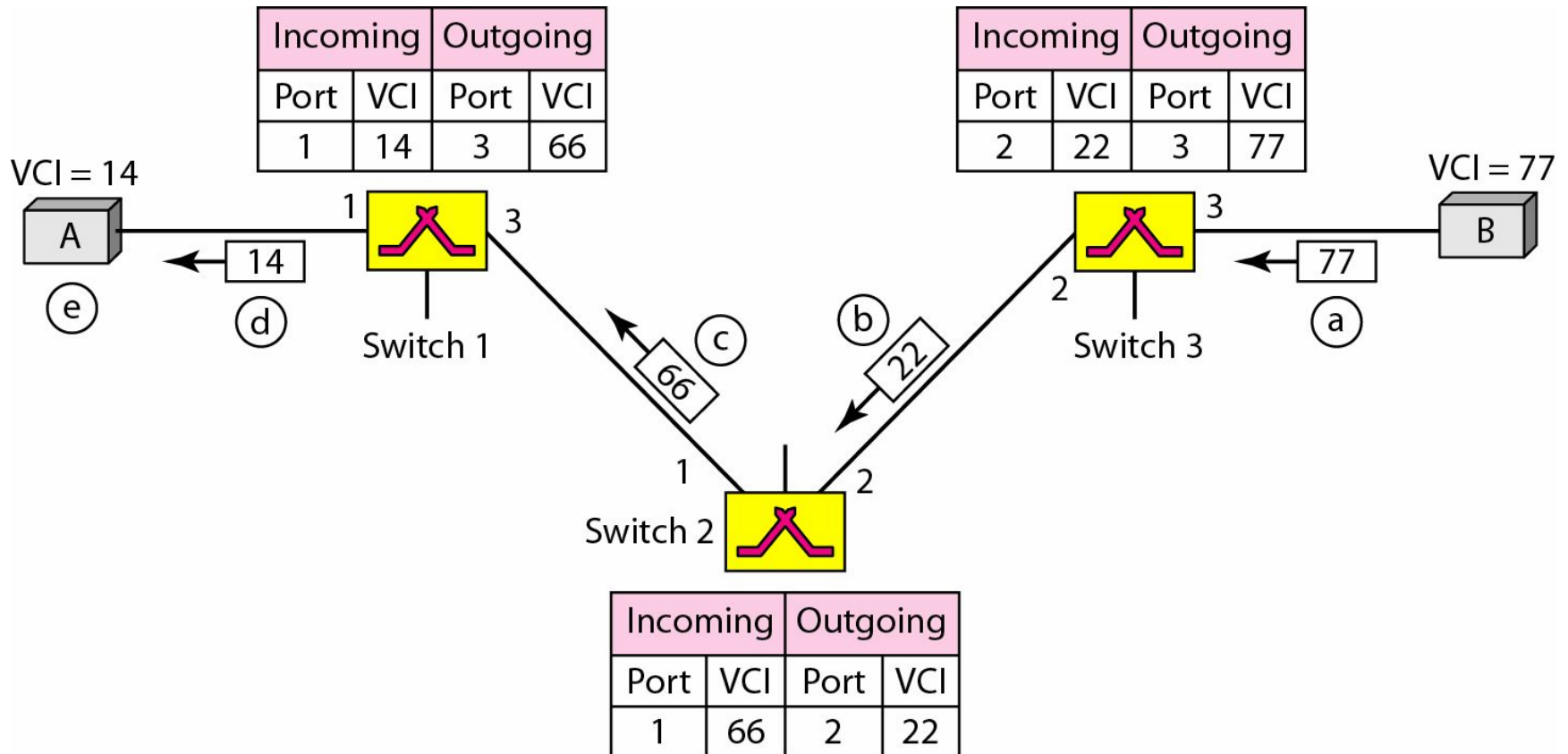
Transmission phases

- Setup phase
- A switch creates an entry for a virtual circuit.
- Request
- acknowledgment

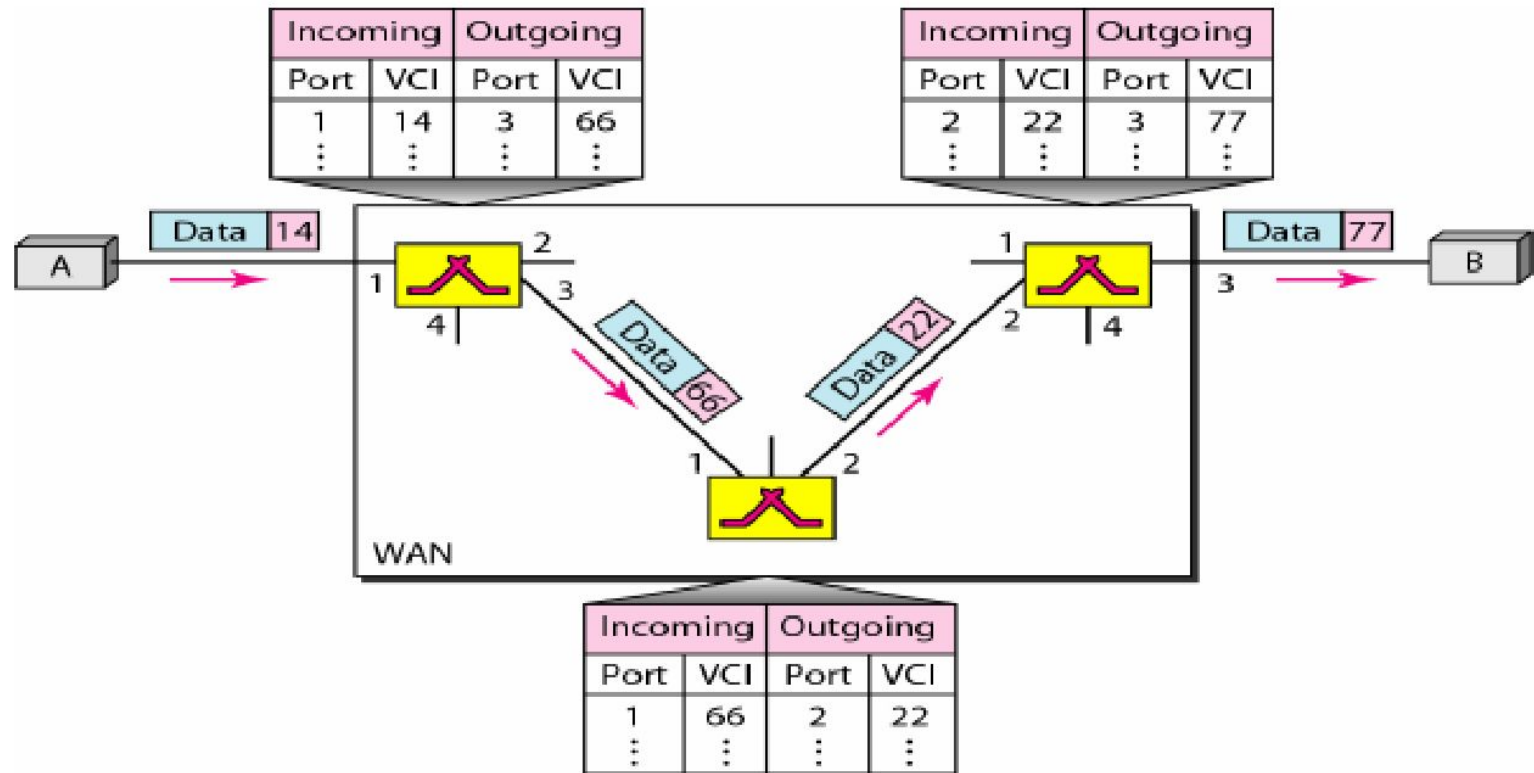
Setup



acknowledgment



Data transfer and teardown phases



After sending all frames, a special frame is send to end the connection

Efficiency

- In virtual-circuit switching, all packets
- belonging to the same source and
- destination travel the same path;
- But the packets may arrive at the destination with different delays if resource allocation is on demand.

Delay

- Total delay = $3T + 3\zeta$ + setup delay + teardown delay

